V-JEPA 2.1 on IntPhys 2 (Main split)

Table 1: Accuracy values showing best model performance for each difficulty level. Rows other than V-JEPA 2.1 are reproduced from IntPhys 2 (Bordes et al., 2025), Table 2: each model's best run per column over a hyperparameter grid, with surprise averaged over time (the paper's default, App. D.1). For V-JEPA 2.1 (shaded, ours), each column is the best of the 6 context lengths; the max-surprise row is given for reference only. Bold: best per model type, ours included.

Model	Type	Easy	Medium	Hard	Overall	Held Out	*IntPhys*
GPT4-o	MLLMs	57.69	54.75	54.17	53.75	53.19	-
Qwen-VL 2.5	MLLMs	50.96	53.25	51.49	52.27	49.12	-
Gemini-1.5 Pro	MLLMs	58.65	53.0	52.67	52.27	52.10	*55.81*
Gemini-2.5 Flash	MLLMs	**64.42**	**56.75**	**54.46**	**55.63**	**56.10**	*56.39*
VideoMAEv2-g	Predictive	46.00	**58.50**	52.73	53.75	53.49	*59.40*
Cosmos-4B	Predictive	46.00	52.00	48.05	49.41	48.84	*85.42*
V-JEPA-h + RoPE	Predictive	52.00	53.00	57.42	53.75	54.65	*98.30*
V-JEPA 2-h	Predictive	54.00	**58.50**	**59.38**	**57.51**	**56.40**	*87.22*
V-JEPA 2.1-g (avg surprise)	Predictive	**63.46**	55.50	55.95	55.14	–	-
max surprise, ref.	Predictive	50.00	52.00	54.76	51.78	–	-
Human	-	96.17	97.8	95.5	96.44	92.44	-

Table 2: Accuracy across property and camera type. For each model, the accuracy of each subset is the best one across a set of hyperparameters (for V-JEPA 2.1: the 6 context lengths, avg surprise; max surprise for reference). The smaller size of each subset contributes to volatility in performance (52 / 68 pairs per Fixed / Moving cell, 94 for Solidity Moving).

	Permanence		Immutability		Continuity		Solidity	
Model	Fixed	Moving	Fixed	Moving	Fixed	Moving	Fixed	Moving
GPT4-o	59.62	**58.82**	58.65	59.56	**54.81**	**57.35**	**56.73**	55.32
Qwen-VL 2.5	53.85	54.41	56.73	53.68	52.88	54.41	50.96	51.06
Gemini-1.5 Pro	55.77	55.88	56.73	56.73	54.80	54.80	56.73	56.73
Gemini-2.5 Flash	**64.42**	**58.82**	**59.62**	**63.97**	**54.81**	55.15	55.77	**56.38**
VideoMAEv2-g	63.46	50.00	54.81	53.69	**65.38**	54.41	48.08	**59.57**
Cosmos-4B	51.92	41.18	50.96	48.32	53.85	50.00	48.08	55.32
V-JEPA-h + RoPE	63.46	**67.65**	51.92	56.38	50.00	57.35	50.00	52.13
V-JEPA 2-h	59.62	57.35	**55.77**	**58.72**	57.69	**75.00**	46.15	58.51
V-JEPA 2.1-g (avg surprise)	**67.31**	54.41	**55.77**	54.41	59.62	60.29	**53.85**	55.32
max surprise, ref.	44.23	55.88	51.92	50.00	59.62	52.94	51.92	53.19
Human	100.0	99.26	97.11	90.44	99.04	94.44	96.15	95.21

Experimental setup. *Model.* Official V-JEPA 2.1 ViT-g/16 release at 384 px from facebookresearch/vjepa2 (checkpoint `vjepa2_1_vitg_384.pt`). The model code is vendored unchanged from that repository, and the context encoder, target (EMA) encoder (1.01 B parameters each) and predictor (55 M) load with strict key matching. V-JEPA 2.1 predicts the concatenation of 4 hierarchical encoder levels; we use only the last one (layer 39), layer-normed as in its training targets, and the predictor's mask token 0 (the one used for video during training).

Surprise. Following Garrido et al. (2025) and the setup reported as best for V-JEPA 2 in IntPhys 2: videos are sampled every 10th frame (6 fps), split into 48-frame windows with stride 2, and for each window the model predicts the features of all frames after a context of $C \in \{12, 18, 24, 30, 36, 42\}$ frames, as in the authors' released V-JEPA 2 config (`vjepa_2.yaml`; the paper lists $\{4, \ldots, 14\}$). The prediction error is the L1 distance between predicted and target features (bf16). With `max_context_mode` (the paper's growing-context protocol), the first window is also predicted from every shorter context $2, 4, \ldots, C-2$, so that early events are not missed. A video's surprise is the average of these errors over time (App. D.1).

Hyperparameters. For V-JEPA and V-JEPA 2 the paper reports, per column, the maximum over window size $\{16, 32, 48\}$ at a fixed 6 fps, prediction length (fixed, or to the end of the window) and context length (App. D.3); V-JEPA 2's best was window 48, prediction to the end of the window, 6 fps (Table 8). We evaluate only that setting and maximise over the 6 context lengths, a subset of their search.

Metric. Relative accuracy on matched pairs: in each scene quadruplet, the impossible video `k_Impossible` must be more surprising than its matched `k_Possible` ($k = 1, 2$), giving 506 pairs on Main (1,012 videos), with 95% confidence intervals of about ± 4.4 points overall and ± 14 on Easy. Easy/Medium/Hard follow the environment lists of the authors' scoring notebook (52/200/168 pairs; the other 86 pairs count only towards Overall). The paper's Easy and Hard values are not multiples of 1/52 and 1/168, so its subsets differ slightly. Held Out was not evaluated.

Observation. With the paper's average surprise, V-JEPA 2.1-g reaches 55.14% overall, below V-JEPA 2-h (57.51%). Max surprise stays at chance (51.78%): for 81–97% of videos the maximum comes from the short-context predictions on the first window, so context length barely matters.

Compute. 6 single-GPU nodes (RTX 5090, 3090, 5070 Ti, 5060 Ti, 4060 Ti, plus one more 16 GB card), ≈21 h wall-clock, 1.3–7.3 min per video depending on the GPU; videos split evenly across ranks, with no inter-node communication.